CS2333

OBJECT ORIENTED PROGRAMMING USING JAVA

MINI PROJECT PRESENTATION

ASSETFLOW

IT Asset Lifecycle & Employee Offboarding Management System

A Java Swing desktop application connected to MySQL

SUBMITTED BY

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The problem we are solving

IT assets move through an employee lifecycle — but records are often scattered.

Scattered records

Employee details, inventory, allocations and returns may be tracked in separate sheets or paper records.

Limited visibility

It can be difficult to identify the current asset holder, check return status or review asset condition.

Exit uncertainty

Teams need to confirm asset return, data handover and access revocation before employee exit.

AssetFlow brings these records into one desktop application backed by a central MySQL database.

Project objectives

Employee records
01

Maintain employee details in a structured database.

Asset inventory
02

Track asset codes, serial numbers, type, condition and status.

Allocation history
03

Record which asset is assigned to which employee and when.

Returns & defects
04

Record returned condition and allow defect reporting and review.

Exit clearance
05

Track asset return, data handover and access-revocation indicators.

Activity visibility
06

Show operational counts and selected activity records.

One system, two user experiences

ADMINISTRATOR WORKSPACE

- Dashboard and operational counts
- Employee management
- IT asset inventory
- Allocations and asset returns
- Defect-report review
- Exit-clearance records
- Activity logs

EMPLOYEE PORTAL

- Personal profile and employee details
- Currently assigned assets
- Allocation and return history
- Submit an asset defect report
- View defect reports
- View exit-clearance information

Role selection at login directs the user to the appropriate interface.

How the application is structured

PRESENTATION LAYER
Java Swing screens • Login • Admin workspace • Employee portal

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APPLICATION & MODEL LAYER
App flow • Person / Employee / Asset models • Operation logic

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DATA ACCESS LAYER
Repository classes • PreparedStatement • Transaction handling

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DATABASE LAYER
JDBC connection • MySQL database • Related records

Swing UI → repository methods → JDBC → MySQL

Technology stack & OOP concepts

Java

Application logic and model classes

Java Swing

Desktop windows, forms, tables and navigation

MySQL

Persistent storage for employee and asset records

JDBC + Connector/J

Java-to-MySQL connectivity

Maven

Dependency and build configuration

Visual Studio Code

Development environment

OBJECT-ORIENTED PROGRAMMING

Classes & objects — Employee, Asset and repository/UI classes.
Encapsulation — private model fields with accessor methods.
Inheritance — Employee extends Person; Swing windows extend JFrame.
Abstraction — repository methods hide SQL and connection details.
Polymorphism — overriding and listener callback implementations.

Core application workflow

LOGIN 01

Choose Admin or Employee and enter credentials.

AUTHENTICATE 02

Check account status, role and password hash.

OPEN WORKSPACE 03

Load the administrator or employee interface.

PERFORM ACTION 04

Use forms and tables to manage records.

SAVE / RETRIEVE 05

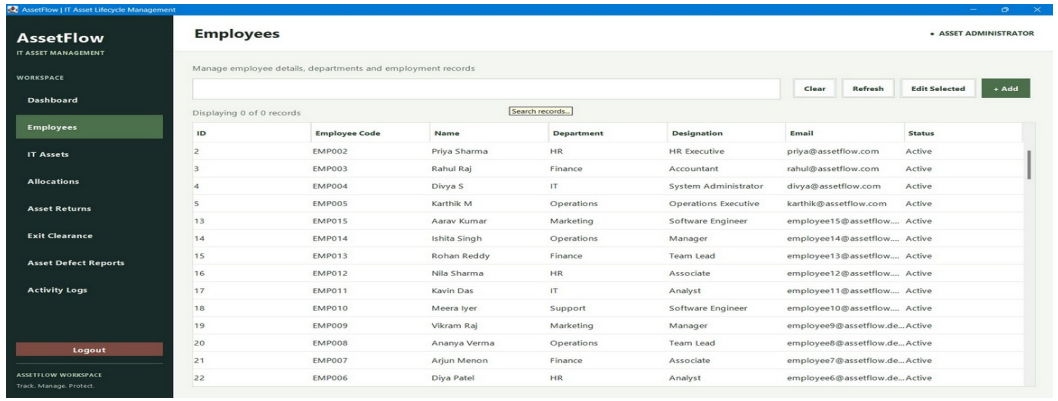
Repositories execute prepared SQL through JDBC.

UPDATE DATABASE 06

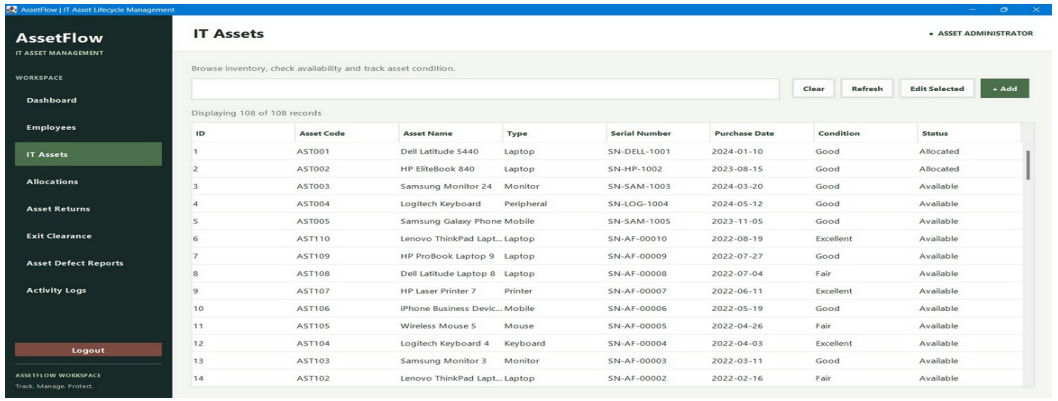
MySQL stores the resulting records and statuses.

Allocation and return operations use database transactions so related updates can be committed together or rolled back on failure.

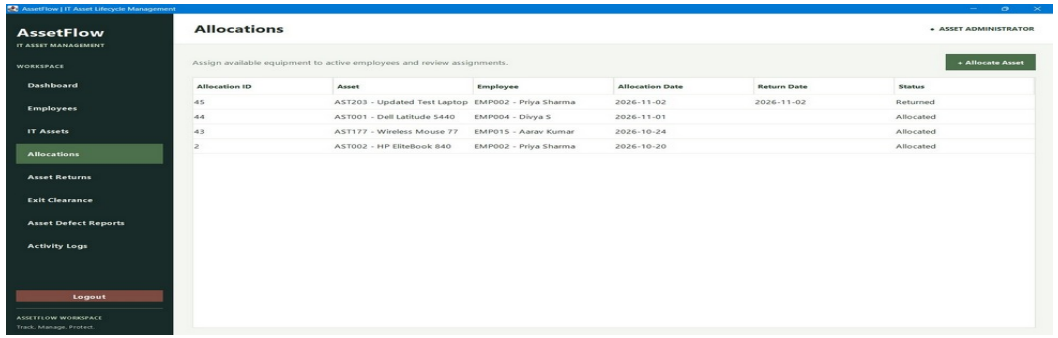
Administrator workspace



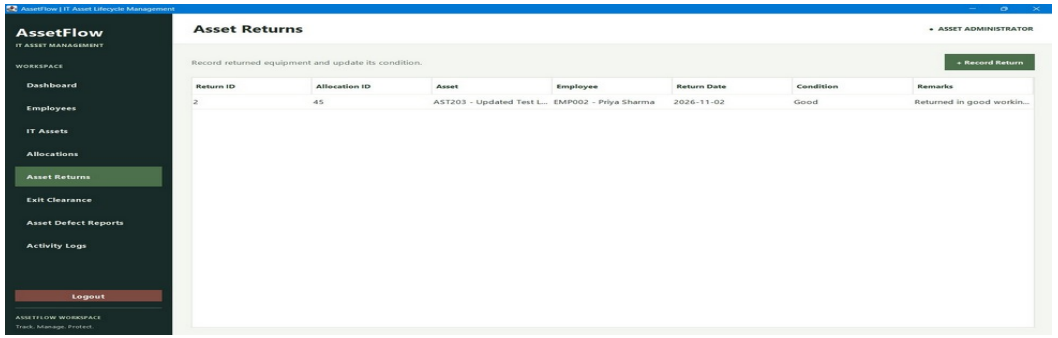
EMPLOYEE MANAGEMENT



ASSET INVENTORY

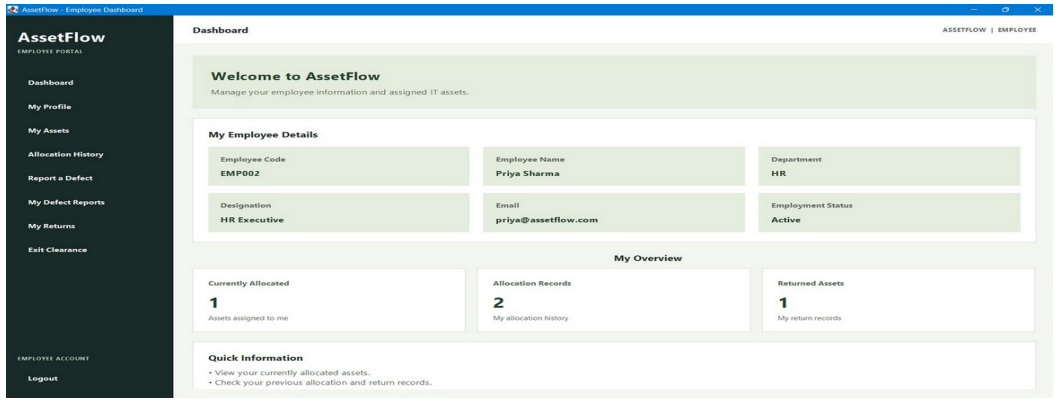


ALLOCATIONS

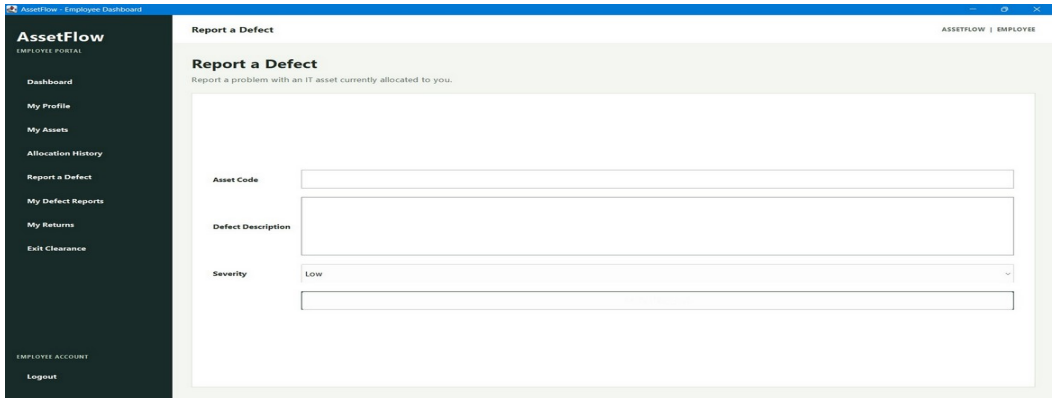


ASSET RETURNS

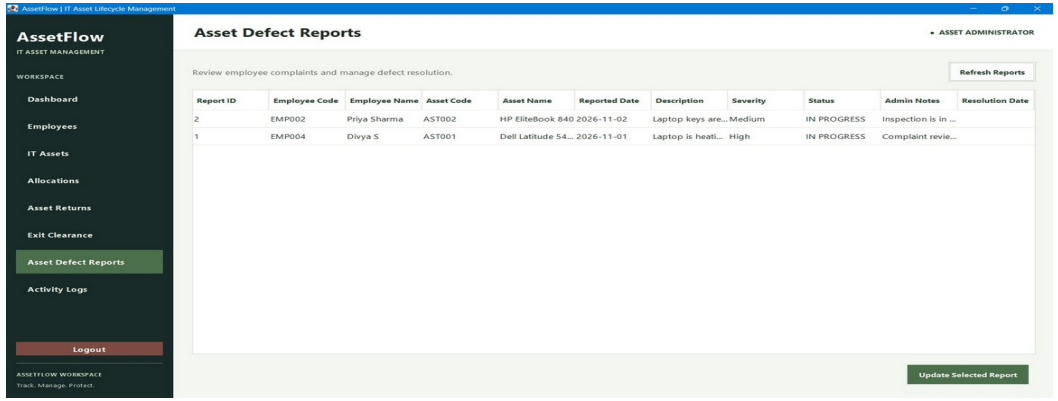
Employee portal & operational tracking



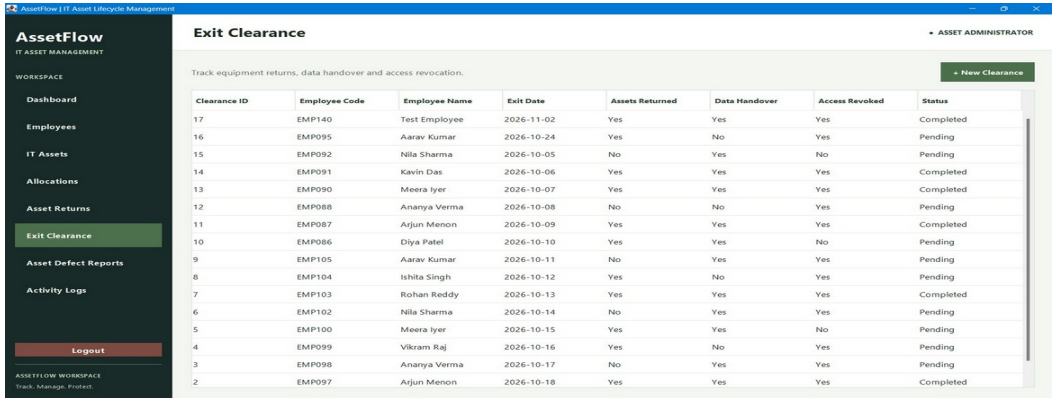
EMPLOYEE PORTAL



DEFECT REPORTING



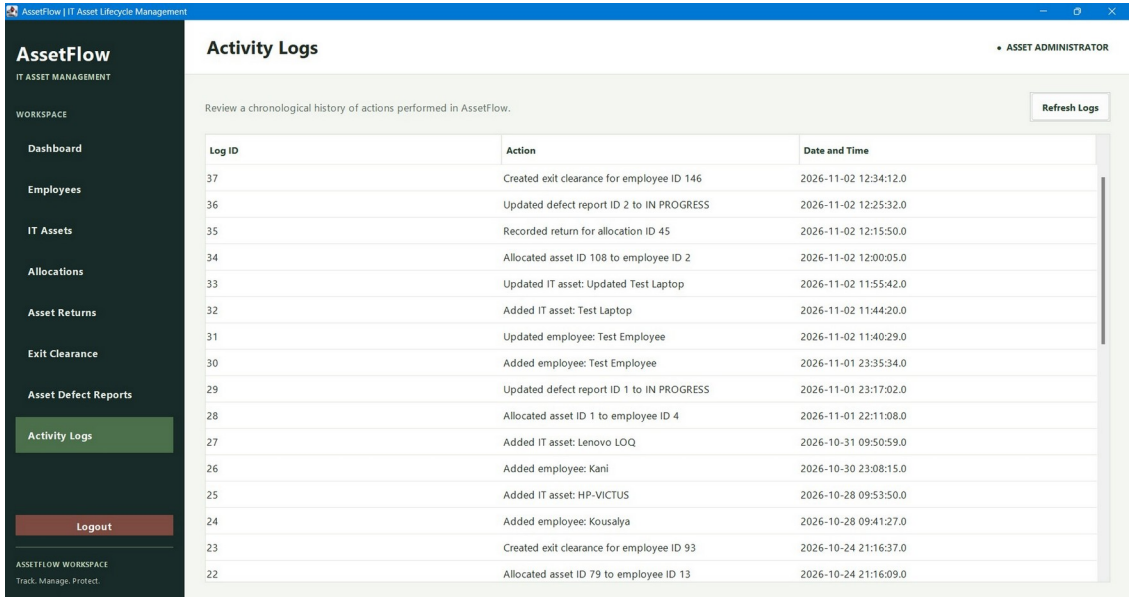
DEFECT TRACKING

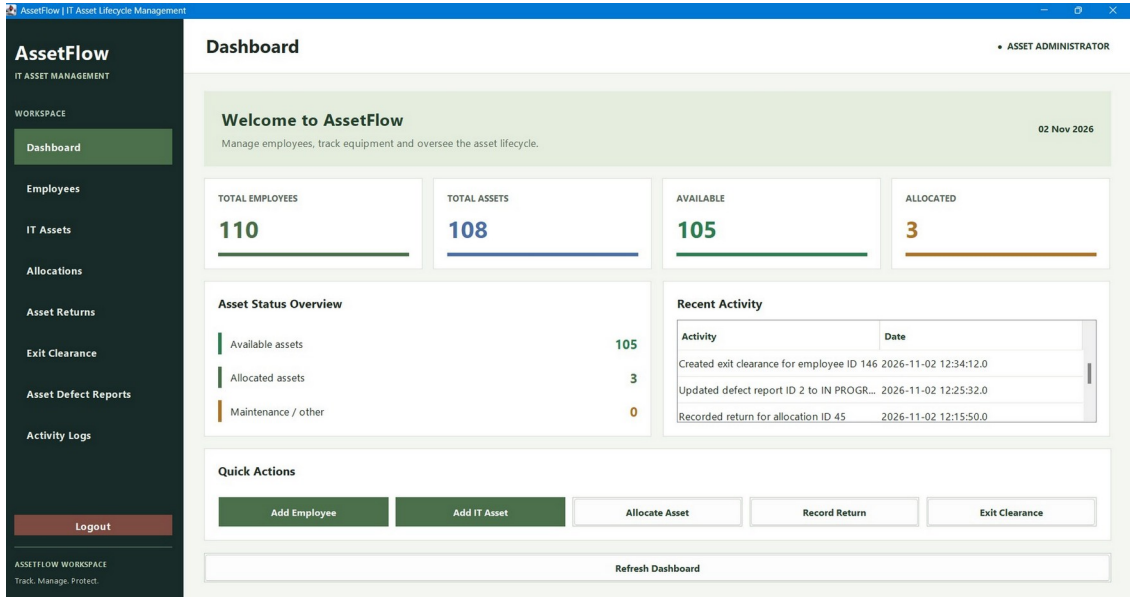


EXIT CLEARANCE

ACTIVITY LOGS ARE ALSO AVAILABLE TO THE ADMINISTRATOR

Activity logs & operational overview





ACTIVITY LOGS

ADMIN DASHBOARD & SUMMARY

The dashboard displays employee and asset counts, asset status overview, recent activity and quick actions.

Database design, outcome & next steps

CORE DATABASE RECORDS

employees • assets • allocations • asset_returns
asset_defect_reports • data_handovers
exit_clearance • user_accounts • activity_logs

KEY RELATIONSHIPS

Employees link to allocations, defect reports, data handovers and exit clearance. Assets link to allocations and defect reports. Returns reference an allocation.

CONCLUSION

AssetFlow brings employee and IT asset records into one Java Swing application backed by MySQL. It demonstrates OOP, JDBC connectivity, relational data design and transaction-based allocation and return updates.

FUTURE ENHANCEMENTS

Dedicated data-handover workflow • Exportable reports • Stronger validation and testing • Notifications • Role-permission refinement • Backup and deployment improvements

References: Oracle Java & Swing documentation • MySQL Reference Manual • MySQL Connector/J Guide • Apache Maven Guides • Supplied AssetFlow source and SQL schema

THANK YOU

ASSETFLOW